TAC-NC Supplementary Proofs

1 Scope of This Supplementary Material

This supplementary material provides detailed proofs of the correctness, convergence, and safety results stated in the main paper, while keeping the notation consistent with the main text.

The proof structure of TAC-NC is organized around three key methods:

1. Method I: constructing tighter arrival curves with BOM and packet-level discretization;
2. Method II: constructing service curves with WOM;
3. Method III: solving the delay vector in cyclic networks by fixed-point iteration.

This supplementary material mainly proves the following claims:

1. the curve produced by Method I still satisfies the definition of an arrival curve and is therefore a valid arrival curve;
2. the curve produced by Method II still satisfies the definition of a service curve and is therefore a valid service curve;
3. the global update in Method III is a contraction mapping under the hyper-period feasibility condition and converges to a unique fixed point;
4. the fixed point produced by Method III yields a safe end-to-end WCD upper bound.

2 Unified Definitions and Assumptions

2.1 Network and Flow Model

Consider a TSN network

$$G = (S \cup E, P, L), \tag{1}$$

where S is the set of switches, E is the set of end systems, $P(v)$ is the set of output ports of switch v, and L is the set of links.

Let F denote the set of flows. For each flow $f \in F$, write

$$f = (src_f, dst_f, T_f, L_f^{\max}, D_f^{ddl}), \tag{2}$$

where T_f is the period, $L_f^{\max}$ is the maximum frame length, and D_f^{ddl} is the deadline. Define further

$$\rho_f \triangleq \frac{L_f^{\max}}{T_f}, \tag{3}$$

which is regarded as an upper bound on the long-term average rate of flow f.

2.2 Basic Network Calculus Concepts

Let $R(t)$ denote the cumulative arrival process, i.e., the total amount of data that has arrived from the current analysis origin up to time t. Let $R^*(t)$ denote the cumulative output process, i.e., the total amount of data output by the system during the same interval.

The analysis origin is taken at the beginning of a busy period. A busy period is a contiguous time interval during which the system continuously has data to process until the queue becomes empty again. After shifting this origin to time 0, we have

$$R(0) = R^*(0) = 0. \tag{4}$$

If a function α satisfies

$$\forall s \leq t, \qquad R(t) - R(s) \leq \alpha(t - s), \tag{5}$$

then α is called a valid arrival curve for the cumulative arrival process $R(t)$.

If a function β satisfies

$$R^*(t) \geq \inf_{0 \leq s \leq t} \{R(s) + \beta(t - s)\}, \tag{6}$$

then β is called a valid service curve for the system output $R^*(t)$.

Throughout this supplementary material, "valid" simply means that the corresponding defining inequality above still holds, so the constructed curve can be used in the subsequent Network Calculus analysis.

The horizontal deviation between α and β is denoted by

$$h(\alpha, \beta) = \sup_{t \geq 0} \inf\{\tau \geq 0 : \alpha(t) \leq \beta(t + \tau)\}. \tag{7}$$

Geometrically, $h(\alpha, \beta)$ is the worst-case minimum right shift required to make the arrival curve α lie everywhere below the service curve β. This horizontal shift corresponds to the worst waiting time experienced by data in the system and therefore serves as a safe delay upper bound in Network Calculus. In the port-level analysis considered here, the port-level delay is given by this horizontal distance between the corresponding arrival and service curves.

2.3 TS-tree, WOM/BOM, and Knee-Point Iteration

For each output port, the combined shaper configuration is represented by a TS-tree:

$$T = (V_{ts}, E_{ts}, \phi), \tag{8}$$

where nodes represent basic shapers or schedulers such as CBS, TAS, SP, and RR; parent-child edges represent conditional serial composition; and sibling nodes represent parallel competition.

For any basic shaper q, let

$$\Theta_q = \{\vartheta_q(t, p_i)\}_{i=1}^{m_q} \tag{9}$$

denote the set of its characteristic functions under all feasible phases. Here p_i represents a different starting offset of the same shaping rule on the time axis. For a periodically gated shaper, changing the observation origin changes the initial service profile seen in the first segment, while leaving the underlying rule unchanged. The parameters p_i enumerate these feasible starting offsets.

Based on this, WOM and BOM are defined as

$$\vartheta_{q,W}(t) = \inf_{p_i} \vartheta_q(t, p_i), \qquad \vartheta_{q,B}(t) = \sup_{p_i} \vartheta_q(t, p_i). \tag{10}$$

WOM denotes the worst output mode, and BOM denotes the best output mode. In this paper, WOM is used to construct service curves, while BOM is used to construct previous-hop output constraints and then derive arrival curves.

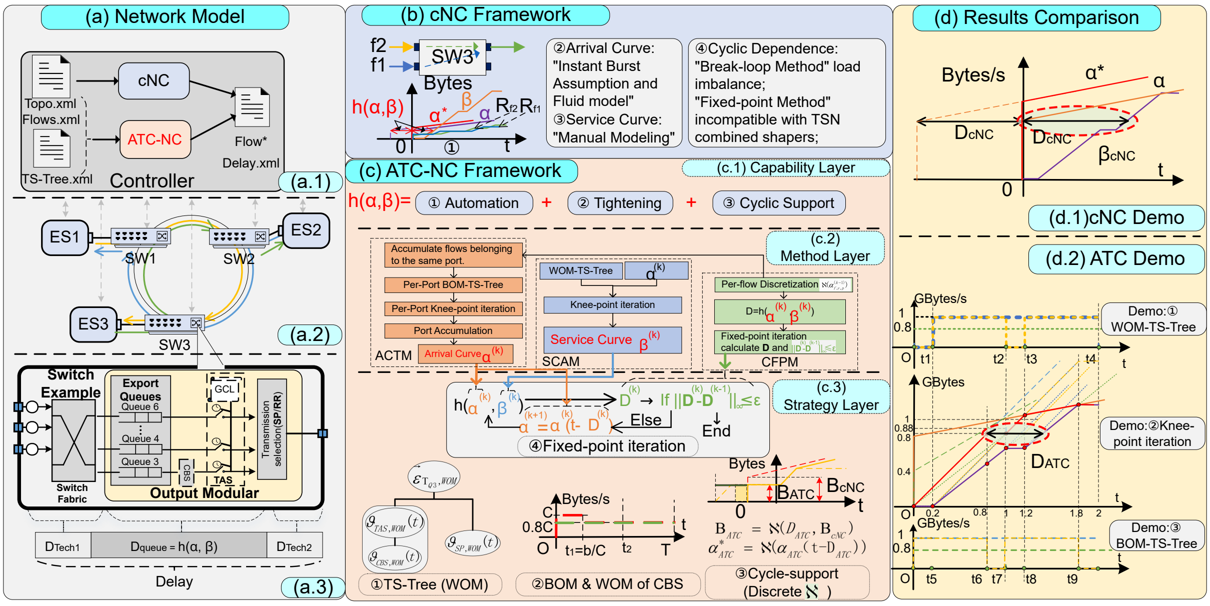

Figure 1: Framework figure from the main paper. The present explanation of TS-tree, WOM/BOM, and knee-point iteration mainly corresponds to (c.3) and (d.2).

Replacing each node in the TS-tree by its WOM envelope yields the WOM-TS-tree

$$T_W \triangleq \mathrm{WOM}(T). \tag{11}$$

Replacing each node in the TS-tree by its BOM envelope yields the BOM-TS-tree

$$T_B \triangleq \mathrm{BOM}(T). \tag{12}$$

Method II generates the port-level service curve on T_W, whereas Method I generates the fastest realizable cumulative output bound for each flow group associated with the same previous-hop output port on T_B.

Given T_W or T_B together with an input arrival curve, TAC-NC uses knee-point iteration to generate the cumulative output curve segment by segment. The knee points correspond to the event points in the main text and mainly include two types: rate-switching instants induced by TS-tree-WOM/BOM, such as TAS gate opening or closing, CBS credit depletion or replenishment, and SP/RR scheduling switches; and instants at which a linear extension from the current fixed knee point along one candidate rate must switch to another feasible branch. In other words, TS-tree-WOM/BOM specifies the feasible output modes and their switching rules, while knee-point iteration fixes the next knee point at those instants and recursively extends the curve.

To facilitate comparison with the main text, Fig. 1 reproduces the framework figure from the main paper. In particular, (c.3) corresponds to TS-tree and WOM/BOM, while (d.2) corresponds to rate states, event points, and the generation of cumulative boundary curves by knee-point iteration.

2.4 Two Classes of Cumulative Curves Generated by TAC-NC

Given an input arrival curve, TAC-NC uses TS-tree-WOM/BOM together with knee-point iteration to generate two classes of cumulative curves:

- the worst cumulative output curve $R_S^*(t)$ on the WOM-TS-tree T_W, which provides a lower bound on the port service capability and further induces the service curve of that port;

- for a group of flows coming from the same previous-hop output port, the fastest realizable cumulative output bound $R^*_{F,u,q\to v,p}(\cdot;\boldsymbol{D})$ on the BOM-TS-tree T_B, which represents the group-level arrival contribution from the previous-hop output port (u,q) to the current port (v,p) and is further used in Method I to form the aggregated arrival curve.

In Method I, per-flow curves are constructed first, and group-level curves are constructed next.

First, for each $f \in F^{u,q}_{v,p}$, let

$$\bar{\alpha}_{f,v,p}(\cdot;\boldsymbol{D}) \tag{13}$$

denote the propagated per-flow arrival curve under the current delay vector $\boldsymbol{D}$, before any BOM processing. Each flow is then discretized:

$$\hat{\alpha}_{f,v,p}(\cdot;\boldsymbol{D}) \triangleq \aleph(\bar{\alpha}_{f,v,p}(\cdot;\boldsymbol{D})). \tag{14}$$

Second, for the current output port (v,p), competing flows may come from multiple previous-hop output ports. We therefore group them by previous-hop output port. Flows from the same previous-hop output port share the same shaping-output constraint and are processed as one group. Define

$$\mathcal{U}_{v,p} \triangleq \{(u,q) \mid \exists f \in F_{v,p},\ \text{prevOut}(f) = (u,q)\} \tag{15}$$

as the set of all previous-hop output ports sending traffic to (v,p). For each $(u,q) \in \mathcal{U}_{v,p}$, define the corresponding group of flows as

$$F^{u,q}_{v,p} \triangleq \{f \in F_{v,p} \mid \text{prevOut}(f) = (u,q)\}. \tag{16}$$

For a fixed previous-hop output port (u,q), the discretized per-flow arrival curves in this group are first summed to obtain the aggregate input

$$\alpha^{sum}_{u,q\to v,p}(\cdot;\boldsymbol{D}) \triangleq \sum_{f\in F^{u,q}_{v,p}} \hat{\alpha}_{f,v,p}(\cdot;\boldsymbol{D}). \tag{17}$$

This aggregate input is then fed into the BOM-TS-tree of the previous-hop output port (u,q), and knee-point iteration is used to generate the fastest cumulative output bound of this group at that port:

$$R^*_{F,u,q\to v,p}(\cdot;\boldsymbol{D}). \tag{18}$$

Accordingly, the arrival contribution of this group to the current port (v,p) is written as

$$\alpha^{grp}_{u,q\to v,p}(\cdot;\boldsymbol{D}) \triangleq R^*_{F,u,q\to v,p}(\cdot;\boldsymbol{D}). \tag{19}$$

Here $R^*_{F,u,q\to v,p}$ denotes the fastest cumulative output produced by this group under the BOM-TS-tree of the previous-hop output port (u,q). Since the flows in this group share the same previous-hop output port and the same shaping rules, TAC-NC uses it as the group-level arrival curve from (u,q) to the current port (v,p).

Finally, the total aggregated arrival curve seen by port (v,p) is

$$\alpha_{agg,v,p}(\cdot;\boldsymbol{D}) = \sum_{(u,q)\in\mathcal{U}_{v,p}} \alpha^{grp}_{u,q\to v,p}(\cdot;\boldsymbol{D}). \tag{20}$$

In short,

$$\alpha_{agg,v,p}(\cdot;\boldsymbol{D}) = \mathcal{A}_{v,p}(\boldsymbol{D}). \tag{21}$$

2.5 Cyclic Updates and Discretization

We first distinguish three notational objects that recur throughout the remainder of the supplementary material.

For any output port (v, p), let $D_{v,p}$ denote the port-level delay component associated with that port. It is a scalar representing the delay at that port.

Collecting the port-level delay components of all analyzed output ports yields the global delay vector

$$\boldsymbol{D} = \left[D_{v_1,p_1}, D_{v_2,p_2}, \ldots, D_{v_m,p_m}\right]^T. \tag{22}$$

$\boldsymbol{D}$ is not the delay of a single port; it is the vector formed by all relevant port delays in the network.

In what follows, $\boldsymbol{D}$ always denotes a finite nonnegative global delay vector.

Under this convention, for any output port (v, p), define the port-level update as

$$\Phi_{v,p}(\boldsymbol{D}) \triangleq h(\alpha_{agg,v,p}(\cdot; \boldsymbol{D}), \beta_{v,p}(\cdot; \boldsymbol{D})), \tag{23}$$

where $\alpha_{agg,v,p}(\cdot; \boldsymbol{D}) = \mathcal{A}_{v,p}(\boldsymbol{D})$ is the aggregated arrival curve induced by $\boldsymbol{D}$, and $\beta_{v,p}(\cdot; \boldsymbol{D})$ is the corresponding WOM service curve. Collecting the updates of all output ports yields the global update

$$\Phi(\boldsymbol{D}) = \left[\Phi_{v_1,p_1}(\boldsymbol{D}), \Phi_{v_2,p_2}(\boldsymbol{D}), \ldots, \Phi_{v_m,p_m}(\boldsymbol{D})\right]^T. \tag{24}$$

One global TAC-NC iteration is therefore

$$\boldsymbol{D}^{(k+1)} = \Phi(\boldsymbol{D}^{(k)}). \tag{25}$$

TAC-NC also uses the discretization operator from the main text:

$$\alpha_{\aleph,f}(t) = \inf_{0 \le s \le t} \left\{ \alpha_f(t - s) + \left\lfloor \frac{s}{T_f} \right\rfloor L_f^{\max} \right\}. \tag{26}$$

Let

$$\gamma_f(t) \triangleq \left\lfloor \frac{t}{T_f} \right\rfloor L_f^{\max}, \tag{27}$$

then $\alpha_{\aleph,f} = \alpha_f \otimes \gamma_f$.

Discretization does not change the traffic model itself. Rather, it introduces the packet-level arrival constraint of periodic flows into arrival-curve propagation. For a flow with period T_f and maximum frame length $L_f^{\max}$, the discretized per-flow arrival curve is no larger than the original fluid-model arrival curve and continues to participate in subsequent grouping by previous-hop output port and BOM processing.

2.6 Hyper-Period Feasibility Condition

In periodic gating scenarios such as TAS/Qbv, service curves are generally not of constant slope. Over some short intervals, the port may provide no service at all. Therefore, to determine whether the cyclic fixed-point iteration can converge, it is not sufficient to examine an instantaneous service rate; one must instead consider whether the port provides sufficient cumulative service over a hyper-period.

Define the hyper-period as

$$H = \mathrm{lcm}(\{T_f \mid f \in F\}). \tag{28}$$

Since H is the least common multiple of all flow periods, flow f appears exactly H/T_f times within one hyper-period H, and the corresponding upper bound on its total data volume is

$$L_f^{\max} \cdot \frac{H}{T_f}. \tag{29}$$

Definition: minimum cumulative service over a hyper-period.

For each output port (v, p), let $S_{v,p}(H)$ denote the minimum cumulative service that can be guaranteed by that port over one hyper-period H. In order to compare service curves across different rounds of cyclic iteration, we assume that this quantity serves as a uniform lower bound on the WOM service curve at the hyper-period scale, i.e., for any finite nonnegative delay vector $\boldsymbol{D}$,

$$S_{v,p}(H) \le \inf_{t \ge 0} \left[\beta_{v,p}(t + H; \boldsymbol{D}) - \beta_{v,p}(t; \boldsymbol{D})\right]. \tag{30}$$

For a strict service curve, $S_{v,p}(H)$ is the minimum number of bytes guaranteed to be transmitted within time H.

Condition: hyper-period feasibility.

We require the port to have positive residual service over a hyper-period, namely

$$\gamma_H \triangleq \max_{(v,p)} \frac{\sum_{f \in F_{v,p}} L_f^{\max} \cdot H/T_f}{S_{v,p}(H)} < 1. \tag{31}$$

Equivalently, for each port (v, p), we require

$$\sum_{f \in F_{v,p}} L_f^{\max} \cdot \frac{H}{T_f} < S_{v,p}(H). \tag{32}$$

That is, over one hyper-period, the total amount of data arriving at the port must be strictly smaller than the minimum service guaranteed by that port over the same hyper-period, i.e., **the total arrival volume must be strictly smaller than the total service volume.**

This does not require surplus service at every instant. For Qbv/TAS, the instantaneous service rate may be 0 when the gate is closed. As long as the total service over a hyper-period remains larger than the total arrival volume, fixed-point iteration may still converge.

2.7 Standing Assumptions

The following assumptions are used throughout the proofs below:

1. the route, period, maximum frame length, and TS configuration of each flow are known;
2. each basic TS characteristic function is piecewise constant or piecewise linear over any finite analysis window;
3. all arrival curves fed into Method I and Method II satisfy the definition of an arrival curve;
4. the global delay vectors $\boldsymbol{D}$ considered below are always finite and nonnegative;
5. for any two input curves, if one is pointwise no larger than the other, then the same pointwise order is preserved after per-flow propagation, per-flow discretization, summation by previous-hop output port, BOM processing on each group, and WOM service generation;
6. for each flow f, discretization captures the packet-level arrival constraint of periodic flows; moreover, the long-term average rate of each per-flow arrival curve generated in the TAC-NC workflow does not exceed $\rho_f = L_f^{\max}/T_f$;
7. each WOM service curve satisfies the hyper-period cumulative-service assumption stated above.

3 Local Curve Constructions Produce Valid Network Calculus Curves

3.1 Cumulative Curves Under Different Phases and WOM/BOM

This subsection introduces the two classes of cumulative curves needed in the later proofs.

For the same basic shaper, different initial phases lead to different characteristic functions; integrating these characteristic functions yields different cumulative curves. We mainly use the following two boundary curves obtained by comparing these cumulative curves:

1. taking the pointwise minimum yields the **worst cumulative output curve**, which is later used to construct the service curve;
2. taking the pointwise maximum yields the **fastest cumulative output curve**, which is later used to construct the arrival curve.

Accordingly:

- WOM asks which feasible phase yields the worst output mode;
- BOM asks which feasible phase yields the best output mode.

Constructing the Service Curve with WOM

Method II needs to show that, if WOM is first taken at the level of each basic shaper, then combined along the TS-tree and further processed by knee-point iteration, the resulting worst cumulative output curve remains a valid service curve of the whole port, that is, it still satisfies the defining service-curve inequality. Proposition P1 states this result.

3.2 Method II Produces a Valid Service Curve

Proposition 1 (P1: Method II Produces a Valid Service Curve)**.** *For any output port and its associated TS-tree, if the input arrival curve satisfies the definition of an arrival curve, then the worst cumulative output curve generated by the WOM envelopes and knee-point iteration is a valid service curve of that port, i.e., it still satisfies the Network Calculus definition of a service curve under the given combined-shaper configuration.*

Proof. We show that the worst cumulative output curve obtained under WOM indeed provides a lower bound on the service capability of the port. The proof proceeds in three steps: first at the level of a single basic shaper, then at the level of the whole TS-tree, and finally through the knee-point iteration.

First, WOM preserves a lower-bound interpretation at the level of a single basic shaper. For each basic shaper q, the WOM characteristic function $\vartheta_{q,W}$ is obtained by taking the pointwise infimum over all feasible phase patterns. Therefore, at every instant, $\vartheta_{q,W}$ is no larger than the service that this shaper can actually provide under any feasible phase.

Second, this lower-bound interpretation is preserved after composing the WOM descriptions of all basic shapers according to the TS-tree. The parent-child and sibling relations in the TS-tree merely combine local components under prescribed composition rules. If every local component is replaced by a lower-bound description that does not exceed the true service, then the combined result cannot exceed the true combined service. Hence, the overall service behavior generated by the WOM-TS-tree remains a lower-bound description of the actual combined service.

Third, the piecewise concatenation performed by knee-point iteration does not destroy this lower-bound property. Let $K = \{t_0, t_1, \ldots, t_N\}$ denote the set of knee points generated by the event-driven iteration. On each interval $[t_k, t_{k+1})$, the slope chosen by the algorithm is no larger than the true feasible service slope allowed by WOM, and the curve is linearly extended with that slope. It follows by induction over the intervals that

$$R_S^*(t) \le R^*(t), \qquad \forall t \ge 0. \tag{33}$$

Both cumulative curves start at 0, so the base case is immediate. The induction step follows from the fact that, given the same starting point, linear extension with a no-larger slope cannot exceed the actual cumulative output on that interval.

Now define

$$\beta_{\mathrm{TAC}}(t) \triangleq R_S^*(t). \tag{34}$$

Since $R(0) = 0$, choosing $s = 0$ in the min-plus convolution yields

$$(R \otimes R_S^*)(t) = \inf_{0 \le s \le t} \{R(s) + R_S^*(t-s)\} \le R(0) + R_S^*(t) = R_S^*(t). \tag{35}$$

Combining this with $R_S^*(t) \le R^*(t)$ gives

$$R^*(t) \ge (R \otimes R_S^*)(t), \qquad \forall t \ge 0. \tag{36}$$

Therefore, $\beta_{\mathrm{TAC}} = R_S^*$ is a valid service curve. □

3.3 BOM Output Curves and Discretized Per-Flow Arrival Curves

Constructing the Arrival Curve with BOM

For the same basic shaper q, let $R_q(t)$ be the cumulative curve corresponding to one feasible phase, and let $\{t_i\}$ be the set of all breakpoints of this cumulative curve. These breakpoints include both slope-change locations and segment boundaries.

For any window length $\tau \ge 0$, define

$$f_i(\tau) = R_q(t_i + \tau) - R_q(t_i), \tag{37}$$

which denotes the cumulative output increment over a window of length τ starting from breakpoint t_i.

Define further

$$\alpha_q(\tau) = \sup_i f_i(\tau). \tag{38}$$

For a fixed window length τ, this function compares all possible cumulative increments obtained by taking the breakpoints as window origins and keeps the largest one.

The next lemma shows that this curve still satisfies the defining inequality of an arrival curve.

Lemma 1 (Fastest Cumulative Output Curve Used in BOM-Based Arrival-Curve Construction)**.** *Let $R_q(t)$ be the cumulative curve of the basic shaper q under one feasible phase, and let $\{t_i\}$ be its set of breakpoints. Define*

$$f_i(\tau) = R_q(t_i + \tau) - R_q(t_i), \qquad \alpha_q(\tau) = \sup_i f_i(\tau). \tag{39}$$

Then α_q is a valid arrival curve for the cumulative variation of R_q over any time window, i.e.,

$$R_q(t) - R_q(s) \le \alpha_q(t-s), \qquad \forall s \le t. \tag{40}$$

Proof. Fix a window length $\tau \ge 0$ and define

$$\varphi(s) = R_q(s + \tau) - R_q(s). \tag{41}$$

This function gives the cumulative increment within a window of fixed length τ starting from time s.

Since R_q is piecewise linear and nondecreasing in the generalized sense, $\varphi(s)$ is also piecewise linear. Therefore, over each interval induced by the breakpoint set $\{t_i\}$, the maximum of $\varphi(s)$ is attained at some interval endpoint, and these endpoints all belong to the breakpoint set. Hence, for any fixed τ,

$$\sup_s \varphi(s) = \max_i f_i(\tau) = \alpha_q(\tau). \tag{42}$$

This shows that, for any given window length τ, it suffices to examine the cumulative increments of windows starting at breakpoints in order to find the global maximum.

Now take any $s \leq t$ and let $\tau = t - s$. Then

$$R_q(t) - R_q(s) = \varphi(s) \leq \sup_s \varphi(s) = \alpha_q(t - s). \tag{43}$$

Therefore, for any time window $[s, t]$, its cumulative increment does not exceed $\alpha_q(t - s)$. This is exactly the defining inequality of an arrival curve, so α_q is indeed a valid arrival curve. □

This lemma shows that, when constructing the arrival curve, the fastest cumulative output curve can be used to describe the fastest output that the previous-hop port may generate toward downstream analysis.

Discretized Per-Flow Arrival Curves

Method I also uses the per-flow discretization introduced in the main text. Starting from an existing arrival curve, discretization incorporates the packet-level arrival constraint of periodic flows and avoids the extra artificial burst introduced by the fluid model when delay propagation is shorter than one period.

Lemma 2 (Discretized Per-Flow Arrival Curves)**.** *Let α_f be a valid arrival curve of flow f, and define*

$$\gamma_f(t) = \left\lfloor \frac{t}{T_f} \right\rfloor L_f^{\max}. \tag{44}$$

Then the discretized curve

$$\alpha_{\aleph,f} = \alpha_f \otimes \gamma_f \tag{45}$$

can still be used as the per-flow arrival curve in the subsequent analysis, and moreover

$$\alpha_{\aleph,f}(t) \leq \alpha_f(t), \qquad \forall t \geq 0. \tag{46}$$

Proof. Since α_f is already a valid arrival curve for flow f, discretization merely adds the packet-level arrival constraint of periodic flows to the extra burst induced by delay propagation. Therefore, discretization does not introduce a burst larger than that of the original fluid model and can still be used as the per-flow input to the subsequent grouping and BOM processing.

Moreover, choosing $s = 0$ in the convolution gives

$$\alpha_{\aleph,f}(t) \leq \alpha_f(t) + \gamma_f(0) = \alpha_f(t). \tag{47}$$

Hence, discretization does not enlarge the original fluid-model arrival curve. □

3.4 Method I Produces a Valid Arrival Curve

The construction in Method I consists of three steps: per-flow discretization, aggregation of flows from the same previous-hop output port, and BOM-TS-tree processing at the previous-hop output port. Flows from the same previous-hop output port share the same output port and the same set of shaping rules; they are therefore processed as one group under BOM. The goal is to show that the final curve produced by these steps still satisfies the Network Calculus definition of an arrival curve.

Proposition 2 (P2a: Correctness of the Aggregated Input in Method I)**.** *For any current output port (v, p) and any previous-hop output port $(u, q) \in \mathcal{U}_{v,p}$, if the propagated per-flow curve $\bar{\alpha}_{f,v,p}(\cdot; \boldsymbol{D})$ is a valid arrival curve for every $f \in F^{u,q}_{v,p}$, then the discretized per-flow curve*

$$\hat{\alpha}_{f,v,p}(\cdot; \boldsymbol{D}) = \aleph(\bar{\alpha}_{f,v,p}(\cdot; \boldsymbol{D}))$$

can still be used as the per-flow arrival curve of that flow, and the aggregated input

$$\alpha^{sum}_{u,q \to v,p}(\cdot; \boldsymbol{D}) = \sum_{f \in F^{u,q}_{v,p}} \hat{\alpha}_{f,v,p}(\cdot; \boldsymbol{D})$$

is also a valid arrival curve for the actual traffic of this group entering the current port (v, p) from the previous-hop output port (u, q).

Proof. By Lemma 2, the discretized curve can still be used as the per-flow arrival curve and is no larger than the original propagated per-flow arrival curve. Therefore, for each $f \in F^{u,q}_{v,p}$,

$$\hat{\alpha}_{f,v,p}(\cdot; \boldsymbol{D}) = \aleph(\bar{\alpha}_{f,v,p}(\cdot; \boldsymbol{D})) \tag{48}$$

remains a valid per-flow arrival curve of that flow.

Let $R^{sum}_{u,q \to v,p}$ denote the actual cumulative arrival process of the flow group $F^{u,q}_{v,p}$. Since every flow in the group is constrained by its discretized curve, summing the inequalities flow by flow yields

$$R^{sum}_{u,q \to v,p}(s + t) - R^{sum}_{u,q \to v,p}(s) \leq \sum_{f \in F^{u,q}_{v,p}} \hat{\alpha}_{f,v,p}(t; \boldsymbol{D}) = \alpha^{sum}_{u,q \to v,p}(t; \boldsymbol{D}) \tag{49}$$

for all $s, t \geq 0$. Hence, $\alpha^{sum}_{u,q \to v,p}(\cdot; \boldsymbol{D})$ is a valid arrival curve for the input traffic of that group. □

Proposition 3 (P2b: Method I Produces a Valid Arrival Curve)**.** *For any current output port (v, p) and any previous-hop output port $(u, q) \in \mathcal{U}_{v,p}$, if the aggregated input $\alpha^{sum}_{u,q \to v,p}(\cdot; \boldsymbol{D})$ is a valid arrival curve, and if the fastest cumulative output produced for this flow group by the BOM-TS-tree and knee-point iteration of the previous-hop output port (u, q) is denoted by*

$$R^{*}_{F,u,q \to v,p}(\cdot; \boldsymbol{D}),$$

then it is a valid arrival curve for the actual output process of that group from (u, q) toward (v, p), i.e., it still satisfies the Network Calculus definition of an arrival curve. Consequently, the final aggregated arrival curve $\alpha_{agg,v,p}(\cdot; \boldsymbol{D})$ is also a valid arrival curve.

Proof. It remains to show that, after BOM is applied to the whole TS-tree of the previous-hop output port and knee-point iteration is performed, the resulting fastest cumulative output can still be used as the arrival curve required by the downstream analysis.

Fix a previous-hop output port $(u, q) \in \mathcal{U}_{v,p}$, and let $R^{out}_{u,q \to v,p}$ denote the actual cumulative output of flow group $F^{u,q}_{v,p}$ from (u, q) toward the current port (v, p).

Under BOM, every basic shaper in the TS-tree is replaced by the most favorable output mode among all its feasible modes. Therefore, for the same valid input, the BOM-TS-tree yields a fastest realizable cumulative output bound that is no smaller than the actual feasible output. By Proposition P2a, the group-level input $\alpha^{sum}_{u,q\to v,p}(\cdot;\boldsymbol{D})$ is already known to be a valid arrival curve. Hence, the curve constructed by knee-point iteration,

$$R^*_{F,u,q\to v,p}(\cdot;\boldsymbol{D}), \tag{50}$$

gives the fastest realizable output bound of this group from the previous-hop output port (u,q) toward the current port (v,p).

Therefore, for any window length t and any window origin s, we have

$$R^{out}_{u,q\to v,p}(s+t) - R^{out}_{u,q\to v,p}(s) \le R^*_{F,u,q\to v,p}(t;\boldsymbol{D}) = \alpha^{grp}_{u,q\to v,p}(t;\boldsymbol{D}) \tag{51}$$

for all $s,t \ge 0$. Thus, the arrival contribution associated with each previous-hop output port is itself a valid arrival curve.

The total arrival seen by port (v,p) is the sum of the contributions from all relevant previous-hop output ports:

$$R^{agg}_{v,p}(s+t) - R^{agg}_{v,p}(s) = \sum_{(u,q)\in\mathcal{U}_{v,p}} \Big(R^{out}_{u,q\to v,p}(s+t) - R^{out}_{u,q\to v,p}(s)\Big). \tag{52}$$

Applying the bound to each term and summing gives

$$R^{agg}_{v,p}(s+t) - R^{agg}_{v,p}(s) \le \sum_{(u,q)\in\mathcal{U}_{v,p}} \alpha^{grp}_{u,q\to v,p}(t;\boldsymbol{D}) = \alpha_{agg,v,p}(t;\boldsymbol{D}). \tag{53}$$

Therefore, the aggregated arrival curve produced by Method I is a valid arrival curve. □

Remark 1. Propositions P2a–P2b show that this construction still satisfies the arrival-curve definition: the aggregated arrival curve produced by per-flow discretization, grouping by previous-hop output port, and BOM processing can be used in the subsequent Network Calculus analysis.

4 Whether Cyclic Updates Yield Finite Results and Their Convergence

This section proves that the global update in cyclic settings yields finite results and converges. We first derive a service-curve lower bound independent of the delay vector, then establish a Lipschitz property of the port-level update, and finally combine it with the hyper-period feasibility condition to obtain a global contraction mapping.

4.1 A Uniform Service-Curve Lower Bound from Hyper-Period Service

Since the WOM service curve may vary with the delay vector $\boldsymbol{D}$, the convergence analysis requires a service lower bound independent of $\boldsymbol{D}$. Such a uniform lower bound can be constructed from the hyper-period service amount.

Lemma 3 (A Uniform Service-Curve Lower Bound Implied by Hyper-Period Service)**.** *For each output port (v,p), let*

$$r_{v,p} \triangleq \frac{S_{v,p}(H)}{H}. \tag{54}$$

Then any WOM-generated service curve $\beta_{v,p}(\cdot;\boldsymbol{D})$, as long as $\boldsymbol{D}$ is a finite nonnegative delay vector, satisfies

$$\underline{\beta}_{v,p}(t) \triangleq \big[r_{v,p}(t-H)\big]^+. \tag{55}$$

Proof. By the definition of $S_{v,p}(H)$ and the busy-period normalization, every window of length H contains at least $S_{v,p}(H)$ units of cumulative service. In particular,

$$\beta_{v,p}(H; \boldsymbol{D}) \geq S_{v,p}(H), \qquad \forall \boldsymbol{D} \text{ finite and nonnegative.} \tag{56}$$

Accumulating the same lower bound over consecutive hyper-period windows yields, for any integer $m \geq 1$,

$$\beta_{v,p}(mH; \boldsymbol{D}) \geq mS_{v,p}(H). \tag{57}$$

Now fix any $t \geq 0$ and let $m = \lfloor t/H \rfloor$. Since $\beta_{v,p}$ is nondecreasing in the generalized sense,

$$\beta_{v,p}(t; \boldsymbol{D}) \geq \beta_{v,p}(mH; \boldsymbol{D}) \geq mS_{v,p}(H). \tag{58}$$

If $t < H$, then $\underline{\beta}_{v,p}(t) = 0$. If $t \geq H$, then $m \geq t/H - 1$, and therefore

$$mS_{v,p}(H) \geq \left(\frac{t}{H} - 1\right) S_{v,p}(H) = r_{v,p}(t - H) = \underline{\beta}_{v,p}(t). \tag{59}$$

Hence, for any finite nonnegative delay vector $\boldsymbol{D}$,

$$\beta_{v,p}(t; \boldsymbol{D}) \geq \underline{\beta}_{v,p}(t). \tag{60}$$

□

4.2 Port-Level Update via $h(\alpha, \beta)$ and Its Lipschitz Property

Based on the uniform service lower bound obtained above, the port-level update at each output port (v, p) is written as

$$\Phi_{v,p}(\boldsymbol{D}) = h(\alpha_{agg,v,p}(\cdot; \boldsymbol{D}), \beta_{v,p}(\cdot; \boldsymbol{D})). \tag{61}$$

Proposition P3 gives a linear control of this quantity with respect to the input delay vector.

Proposition 4 (P3: Lipschitz Property of the Port-Level Update)**.** *Under the standing assumptions and the hyper-period feasibility condition, for any two finite nonnegative delay vectors $\boldsymbol{D}_1, \boldsymbol{D}_2$, let*

$$\delta = \|\boldsymbol{D}_2 - \boldsymbol{D}_1\|_\infty. \tag{62}$$

Then for any output port (v, p),

$$|\Phi_{v,p}(\boldsymbol{D}_2) - \Phi_{v,p}(\boldsymbol{D}_1)| \leq \frac{\sum_{f \in F_{v,p}} L_f^{\max} \cdot H/T_f}{S_{v,p}(H)} \delta. \tag{63}$$

Proof. Fix any finite nonnegative delay vector $\boldsymbol{D}$. By P1, P2a, and P2b, the WOM service curve and the aggregated arrival curve computed under this $\boldsymbol{D}$ are both valid. For any output port (v, p), the aggregated arrival curve satisfies

$$\limsup_{t \to \infty} \frac{\alpha_{agg,v,p}(t; \boldsymbol{D})}{t} \leq \sum_{f \in F_{v,p}} \rho_f. \tag{64}$$

The hyper-period feasibility condition gives

$$\sum_{f \in F_{v,p}} \rho_f = \frac{\sum_{f \in F_{v,p}} L_f^{\max} \cdot H/T_f}{H} < \frac{S_{v,p}(H)}{H}. \tag{65}$$

By Lemma 3, $\beta_{v,p}(\cdot; \boldsymbol{D})$ is lower-bounded by the uniform service curve $\underline{\beta}_{v,p}(t) = [r_{v,p}(t - H)]^+$, where $r_{v,p} = S_{v,p}(H)/H$. Therefore,

$$h(\alpha_{agg,v,p}(\cdot; \boldsymbol{D}), \beta_{v,p}(\cdot; \boldsymbol{D})) < \infty, \tag{66}$$

so $\Phi_{v,p}(\boldsymbol{D})$ is well defined for every finite nonnegative $\boldsymbol{D}$.

We next estimate the difference induced by two inputs. Let

$$\delta = \|\boldsymbol{D}_2 - \boldsymbol{D}_1\|_\infty. \tag{67}$$

For any flow f, if the delay parameter propagated to the current port changes by at most δ, then the corresponding shifted arrival curve changes by at most $\rho_f\delta$ at every time. After per-flow discretization, grouping by previous-hop output port, summation within each group, and BOM processing, this variation is still bounded by the sum of the long-term average rates of the relevant flows multiplied by δ. Hence, for any port (v,p),

$$\alpha_{agg,v,p}(t;\boldsymbol{D}_2) \le \alpha_{agg,v,p}(t;\boldsymbol{D}_1) \; + \left(\sum_{f\in F_{v,p}} \rho_f\right)\delta, \qquad \forall t \ge 0. \tag{68}$$

Define

$$\Delta_{v,p} \triangleq \left(\sum_{f\in F_{v,p}} \rho_f\right)\delta = \frac{\sum_{f\in F_{v,p}} L_f^{\max}\cdot H/T_f}{H}\,\delta. \tag{69}$$

That is, when the input changes from $\boldsymbol{D}_1$ to $\boldsymbol{D}_2$, the aggregated arrival curve at port (v,p) can increase by at most $\Delta_{v,p}$.

We then estimate the change of $h(\alpha,\beta)$. By Lemma 3, any WOM service curve at port (v,p) satisfies

$$\underline{\beta}_{v,p}(t) = \left[r_{v,p}(t-H)\right]^+, \qquad r_{v,p} = \frac{S_{v,p}(H)}{H}. \tag{70}$$

If the arrival curve increases by at most $\Delta_{v,p}$, then the additional horizontal shift required to restore the comparison relation is at most

$$\frac{\Delta_{v,p}}{r_{v,p}}. \tag{71}$$

Applying the same argument after swapping $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ yields the absolute-value estimate

$$|\Phi_{v,p}(\boldsymbol{D}_2) - \Phi_{v,p}(\boldsymbol{D}_1)| \le \frac{\Delta_{v,p}}{r_{v,p}} = \frac{\sum_{f\in F_{v,p}} L_f^{\max}\cdot H/T_f}{S_{v,p}(H)}\,\delta. \tag{72}$$

This proves the proposition. □

4.3 Contraction and Convergence Under the Hyper-Period Feasibility Condition

The port-level Lipschitz estimate yields the contraction of the global update Φ, and hence the uniqueness of the fixed point and the convergence of the iteration.

Theorem 1 (P4: Convergence Under the Hyper-Period Feasibility Condition)**.** *If the hyper-period feasibility condition holds, then the global update Φ of TAC-NC is a contraction mapping under the infinity norm. Therefore, the iteration*

$$\boldsymbol{D}^{(k+1)} = \Phi(\boldsymbol{D}^{(k)}) \tag{73}$$

admits a unique fixed point and converges to it from any finite nonnegative initial value.

Proof. By Proposition P3, for every output port (v,p),

$$|\Phi_{v,p}(\boldsymbol{D}_2) - \Phi_{v,p}(\boldsymbol{D}_1)| \le \frac{\sum_{f\in F_{v,p}} L_f^{\max}\cdot H/T_f}{S_{v,p}(H)}\,\|\boldsymbol{D}_2 - \boldsymbol{D}_1\|_\infty. \tag{74}$$

Taking the maximum over all ports gives

$$\|\Phi(\boldsymbol{D}_2) - \Phi(\boldsymbol{D}_1)\|_\infty \le \gamma_H \|\boldsymbol{D}_2 - \boldsymbol{D}_1\|_\infty, \tag{75}$$

where

$$\gamma_H = \max_{(v,p)} \frac{\sum_{f\in F_{v,p}} L_f^{\max} \cdot H/T_f}{S_{v,p}(H)}. \tag{76}$$

The hyper-period feasibility condition states exactly that, for every port,

$$\sum_{f\in F_{v,p}} L_f^{\max} \cdot \frac{H}{T_f} < S_{v,p}(H), \tag{77}$$

and therefore

$$\gamma_H < 1. \tag{78}$$

Hence Φ is a contraction mapping under the infinity norm. By the Banach fixed-point theorem, Φ admits a unique fixed point, and the iteration

$$\boldsymbol{D}^{(k+1)} = \Phi(\boldsymbol{D}^{(k)}) \tag{79}$$

converges to that fixed point from any finite nonnegative initial value. □

Corollary 1 (Existence and Uniqueness of the Fixed Point)**.** *Under the conditions of Theorem P4, the cyclic update of TAC-NC admits a unique fixed point* $\boldsymbol{D}^*$.

5 Safety of the Converged Delay Vector

Section 4 has shown that the cyclic update converges to a unique fixed point. We now prove that the fixed point produced by Method III is a safe end-to-end WCD upper bound.

Theorem 2 (P5: The Fixed Point Produced by Method III Is a Safe End-to-End WCD Upper Bound)**.** *Under the standing assumptions and the hyper-period feasibility condition, the fixed point* $\boldsymbol{D}^*$ *produced by Method III gives an end-to-end WCD upper bound for every flow. In particular, for any flow* f,

$$d_f^{actual} \le \sum_{(v,p)\in R_f} D^*_{v,p}. \tag{80}$$

Proof. We use induction to show that the port-level delay components obtained in every iteration are safe upper bounds.

In the initial round, the periodic-flow arrival curves provided by the source end systems upper-bound the actual arrival processes. Assume that in round $k-1$, the per-flow arrival curves propagated to each previous-hop output port upper-bound the actual output processes. Consider any current output port (v,p) in round k. Method I first discretizes the per-flow arrival curves. By Lemma 2, discretization does not enlarge the original arrival curve and still preserves the packet-level arrival constraint of periodic flows. The flows from the same previous-hop output port are then summed within the same group, and the resulting sum upper-bounds the actual input of that group. Finally, Proposition P2b shows that the BOM-TS-tree of the previous-hop output port provides the fastest cumulative output toward the current port for that group. After summing the arrival contributions from all previous-hop ports, the resulting $\alpha_{agg,v,p}(\cdot;\boldsymbol{D}^{(k-1)})$ upper-bounds the actual aggregated arrival process at the current port.

On the other hand, by P1, the service curve $\beta_{v,p}(\cdot;\boldsymbol{D}^{(k-1)})$ used in this round is valid. Therefore, the standard Network Calculus delay theorem gives

$$d_{v,p}^{actual} \le h(\alpha_{agg,v,p}(\cdot;\boldsymbol{D}^{(k-1)}), \beta_{v,p}(\cdot;\boldsymbol{D}^{(k-1)})) = D_{v,p}^{(k)}. \tag{81}$$

Hence, the port-level delay component computed in every round is a safe upper bound on the corresponding actual port-level delay.

By Theorem P4, the iteration $\boldsymbol{D}^{(k+1)} = \Phi(\boldsymbol{D}^{(k)})$ converges to the unique fixed point $\boldsymbol{D}^*$. Since the upper-bound relation holds in every round, it is preserved in the limit, and each $D^*_{v,p}$ remains a safe upper bound on the corresponding actual port-level delay.

For any flow f with route R_f, its actual end-to-end delay is no larger than the sum of the actual port-level delays along its route. Therefore,

$$d_f^{actual} \leq \sum_{(v,p)\in R_f} D^*_{v,p}. \tag{82}$$

Thus, the fixed point $\boldsymbol{D}^*$ produced by Method III is an end-to-end WCD upper bound. □

6 Comparison with Classical cNC Under the Same Port-Level Service Assumption

This section gives the comparison result using notation consistent with the main text. If the two methods use the same port-level service assumption and the arrival curve constructed by TAC-NC is pointwise no larger than that constructed by classical cNC, then the monotonicity of $h(\alpha, \beta)$ in α implies that the port-level delay given by TAC-NC is no larger than that given by classical cNC.

Theorem 3 (P6: Relative Tightness with Respect to Classical cNC Under the Same Port-Level Service Assumption)**.** *Let Φ^{cNC} denote a comparison update defined over the same finite nonnegative delay range as TAC-NC. It uses the same family of port-level service curves $\beta_{v,p}(\cdot;\boldsymbol{D})$ as TAC-NC, but replaces the refined arrival-curve construction of TAC-NC by the classical cNC arrival-curve construction $\alpha^{\mathrm{cNC}}_{agg,v,p}(\cdot;\boldsymbol{D})$. Assume that:*

1. *for any finite nonnegative delay vector $\boldsymbol{D}$ and any output port (v,p),*

$$\alpha_{agg,v,p}(\cdot;\boldsymbol{D}) \leq \alpha^{\mathrm{cNC}}_{agg,v,p}(\cdot;\boldsymbol{D}) \tag{83}$$

   holds pointwise;

2. *the comparison update*

$$\Phi^{\mathrm{cNC}}_{v,p}(\boldsymbol{D}) \triangleq h(\alpha^{\mathrm{cNC}}_{agg,v,p}(\cdot;\boldsymbol{D}), \beta_{v,p}(\cdot;\boldsymbol{D})) \tag{84}$$

   yields finite results in every round over the same finite nonnegative delay range and remains monotone;

3. *the TAC-NC iteration and the cNC comparison iteration start from the same finite nonnegative initial value and converge to the fixed points $\boldsymbol{D}^*$ and $\boldsymbol{D}^{\mathrm{cNC},*}$, respectively.*

Then

$$\boldsymbol{D}^* \leq \boldsymbol{D}^{\mathrm{cNC},*} \tag{85}$$

holds componentwise. Consequently, for any flow f,

$$\bar{D}_f^{\mathrm{TAC}} \triangleq \sum_{(v,p)\in R_f} D^*_{v,p} \leq \sum_{(v,p)\in R_f} D^{\mathrm{cNC},*}_{v,p} \triangleq \bar{D}_f^{\mathrm{cNC}}. \tag{86}$$

Proof. Fix any finite nonnegative delay vector $\boldsymbol{D}$. By assumption, for each output port (v,p),

$$\alpha_{agg,v,p}(\cdot;\boldsymbol{D}) \leq \alpha^{\mathrm{cNC}}_{agg,v,p}(\cdot;\boldsymbol{D}) \tag{87}$$

holds pointwise, and both sides use the same port-level service curve. Since, for fixed β, the horizontal deviation $h(\alpha, \beta)$ is monotonically increasing in α, it follows that

$$\Phi_{v,p}(\boldsymbol{D}) = h(\alpha_{agg,v,p}(\cdot; \boldsymbol{D}), \beta_{v,p}(\cdot; \boldsymbol{D})) \le h(\alpha^{\mathrm{cNC}}_{agg,v,p}(\cdot; \boldsymbol{D}), \beta_{v,p}(\cdot; \boldsymbol{D})) = \Phi^{\mathrm{cNC}}_{v,p}(\boldsymbol{D}). \tag{88}$$

Hence, for every such $\boldsymbol{D}$,

$$\Phi(\boldsymbol{D}) \le \Phi^{\mathrm{cNC}}(\boldsymbol{D}) \tag{89}$$

holds componentwise.

Let both iterations start from the same finite nonnegative initial value:

$$\boldsymbol{D}^{(0)} = \boldsymbol{D}^{(0)}_{\mathrm{cNC}}. \tag{90}$$

We prove by induction that

$$\boldsymbol{D}^{(k)} \le \boldsymbol{D}^{(k)}_{\mathrm{cNC}}, \qquad \forall k \ge 0. \tag{91}$$

The base case is immediate. If the claim holds at round k, then using the monotonicity of Φ and Φ^{cNC} yields

$$\boldsymbol{D}^{(k+1)} = \Phi(\boldsymbol{D}^{(k)}) \le \Phi(\boldsymbol{D}^{(k)}_{\mathrm{cNC}}) \le \Phi^{\mathrm{cNC}}(\boldsymbol{D}^{(k)}_{\mathrm{cNC}}) = \boldsymbol{D}^{(k+1)}_{\mathrm{cNC}}. \tag{92}$$

Thus the induction is complete.

By assumption, the two sequences converge to the fixed points $\boldsymbol{D}^*$ and $\boldsymbol{D}^{\mathrm{cNC},*}$, respectively. Taking the limit in the componentwise inequality above gives

$$\boldsymbol{D}^* \le \boldsymbol{D}^{\mathrm{cNC},*}. \tag{93}$$

Summing along the route R_f of any flow f yields

$$\bar{D}^{\mathrm{TAC}}_f = \sum_{(v,p)\in R_f} D^*_{v,p} \le \sum_{(v,p)\in R_f} D^{\mathrm{cNC},*}_{v,p} = \bar{D}^{\mathrm{cNC}}_f. \tag{94}$$

This proves the theorem. □